Day 38 EBIKE101 Afternoon Lab — Homunculous Finish-Work Diagnosis

Program: CTS Cohort 6

Course: EBIKE101

Date: Tuesday, 2026-09-08

Lab type: Live project diagnosis and approved finish work

Lab objective

Students will diagnose the current state of the E-bike Homunculous, identify what is required to finish it, and begin only the safe work that has been verified and approved.

Instructor setup

Before students begin:

1. Place the bike where all sides can be accessed.
2. Keep battery connection under instructor control.
3. Remove or isolate any obviously unsafe condition.
4. Stage hand tools, meter, labels, zip ties, and worksheet.
5. Define the boundary: diagnosis first, power later only if earned by evidence.

Lab roles

Assign or rotate:

- Mechanical lead: checks frame, wheels, brakes, fasteners, chain path.
- Electrical tracer: maps source, protection, controller, controls, motor.
- Documentation lead: worksheet, board, photo evidence, parts list.
- Safety observer: watches pinch points, battery boundary, tool use, moving parts.
- Parts lead: organizes missing parts, hardware, adapters, labels.

Part 1 — Mechanical readiness inspection

Students inspect:

- Wheel security
- Axle nut condition
- Torque arm or anti-rotation support if applicable
- Brake function
- Tire condition and pressure
- Chain alignment
- Crank/pedal condition
- Sharp edges
- Cable rub points
- Loose fasteners
- Battery/controller mounting

Stop if:

- Motor wheel can shift in dropout
- Brakes cannot stop the wheel
- Chain path is unsafe
- Battery mount is loose
- Any wire can reach tire, chain, brake rotor, or crank

Part 2 — Electrical/control inventory

Students identify and document:

- Battery voltage/rating if labeled
- Controller label/rating if visible
- Main power connector
- Fuse/protection location
- Display connector
- PAS connector/sensor/magnet ring
- Throttle presence or absence
- Brake cutoff connectors
- Motor cable
- Accessory wires
- Damaged connectors
- Unknown wires

Rule:

Unknown wires stay unknown until traced, labeled, or verified. Guessing does not become truth just because someone says it confidently in a shop voice.

Part 3 — Diagnostic decision tree

Use this order:

1. Is the bike mechanically safe to inspect?
2. Is the battery appropriate and controlled?
3. Is there protection between battery and controller?
4. Does the controller have a secure mount?
5. Are control inputs present?
6. Are brake cutoffs present and not stuck active?
7. Is the PAS sensor installed and aligned?
8. Is the throttle present, missing, or optional for this controller?
9. Is the motor cable undamaged and supported?
10. What must be fixed before first power-up?

Part 4 — Approved finish work

Allowed student work may include:

- Brake adjustment

- Tightening non-critical hardware under supervision
- Labeling wires
- Improving cable routing
- Adding strain relief
- Cleaning mounting surfaces
- Organizing connectors
- Creating a parts-needed list
- Documenting evidence
- Non-powered meter checks where appropriate

Instructor-only or instructor-controlled work:

- Battery connection
- First power-up
- Controller testing
- Motor spin testing
- Fuse replacement if cause is unknown
- Any modification to high-current wiring
- Any ride test

Part 5 — First power-up gate

Do not perform a powered test unless all are checked:

- Mechanical assembly secure
- Brakes functional
- Drive wheel controlled
- Battery voltage/rating appropriate
- Fuse/protection path verified
- Controller mounted
- No melted/damaged wires
- No unrestrained motor cables
- No hands near moving parts
- Instructor gives explicit approval

Completion target

The lab is complete when the team can answer:

1. What works or appears complete?
2. What is missing?
3. What is unsafe?
4. What is unknown?
5. What should be done next?

Finishing the bike is excellent. Finishing the diagnosis is required.